Southwire®
POWER GENERATION SOLUTIONS

POWER GENERATION SOLUTIONS

WIRE & CABLE FOR GAS POWER

GAS POWER CAMPUS · FROM THE GRID TO THE STACK

ACCELERATING TIME TO POWER™





ONE CAMPUS.
EVERY CABLE.

A 3x1 combined-cycle plant with carbon capture, battery storage, a modular power yard, an LNG terminal and a behind-the-meter data centre: one campus, and from the 230 kV grid connection to the last instrument loop, all of it runs on cable. This brochure follows that cable through the plant.

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THE PLANT AT A GLANCE

WHERE THE CABLE GOES



SK-3X1 CAMPUS PLAN, DRAWN FROM THE 3D MODEL · NORTH UP

01 GRID CONNECTION
02 POWER ISLAND
03 CARBON CAPTURE
04 BATTERY STORAGE
05 MODULAR & TEMPORARY POWER

BY THE NUMBERS

4.6 MI
of cable tray routes on the racks and in the halls

4.9 MI
of duct-bank and buried cable routes

4,946 FT
of 230 kV underground cable routes

418
pieces of equipment with their own cable applications

Route lengths measured along the modelled tray, duct-bank and cable routes (not cable footage); data-centre campus excluded.

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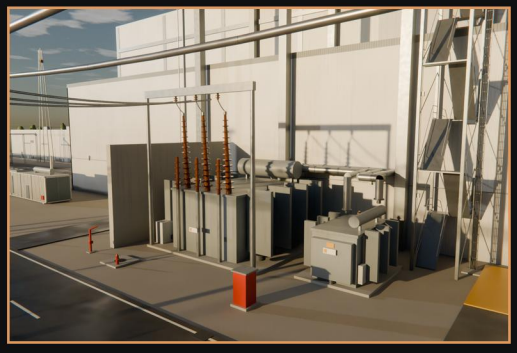
230 kV TO THE GRID

The plant's output leaves through four generator step-up transformers and a breaker-and-a-half 230 kV switchyard. Where the line runs underground, high-voltage XLPE cable rises on cleated risers to sealing ends and surge arresters.

01 HV XLPE TRANSMISSION CABLE
Copper conductor, XLPE insulation, aluminium laminate moisture barrier and HDPE jacket for the buried 230 kV run.

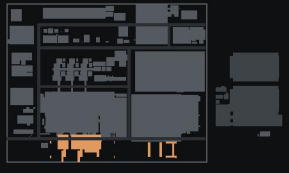
02 FEWER JOINTS
Long continuous lengths keep splices out of the duct bank.

03 ONE SUPPLIER
Cable, accessories and field support for the complete circuit.



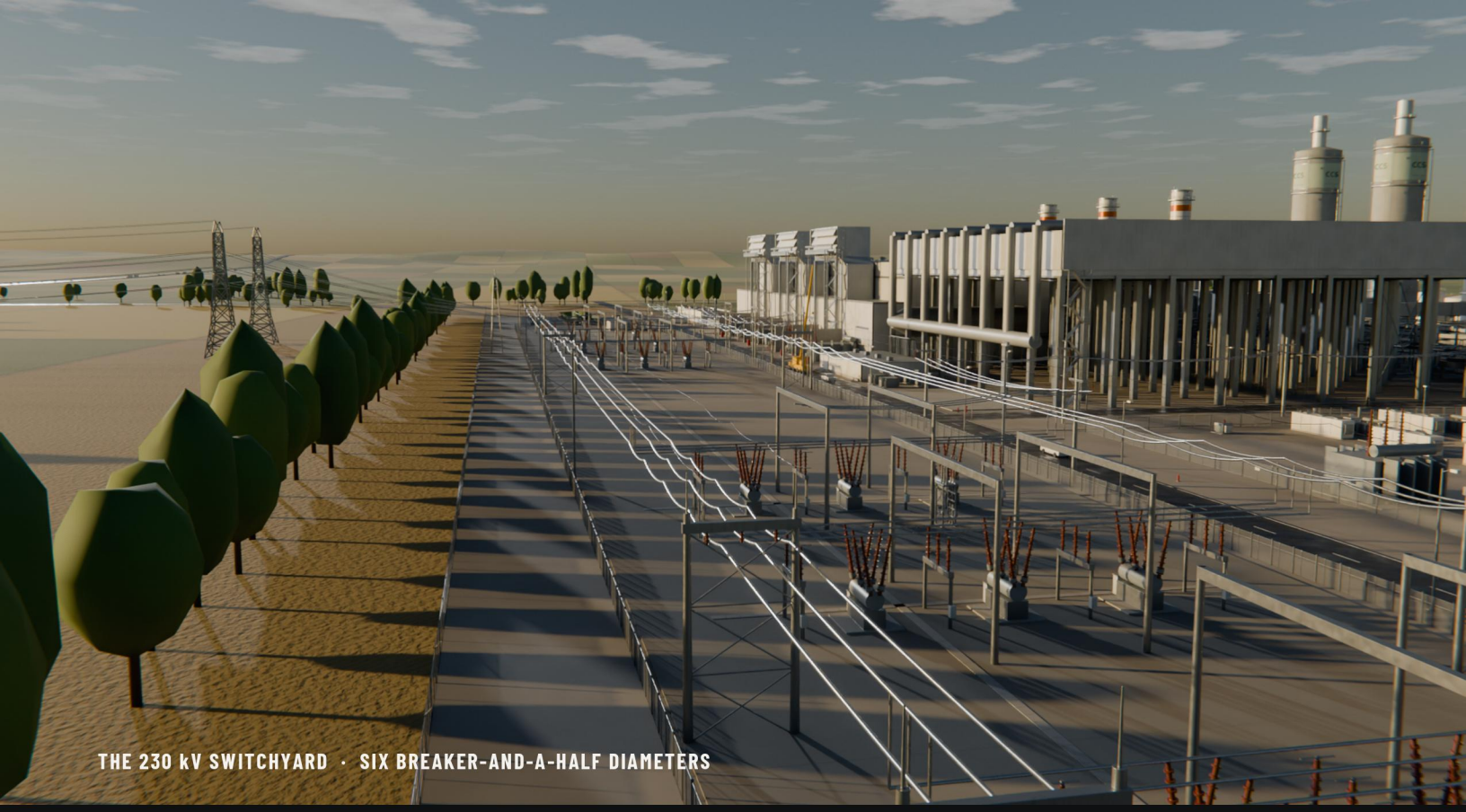
Generator step-up transformers facing the switchyard

230 kV | 2500 KCMIL CU | XLPE | AL LAMINATE | ICEA S-108-720 | 4/0 BARE CU GROUND



WHERE IN THE PLANT

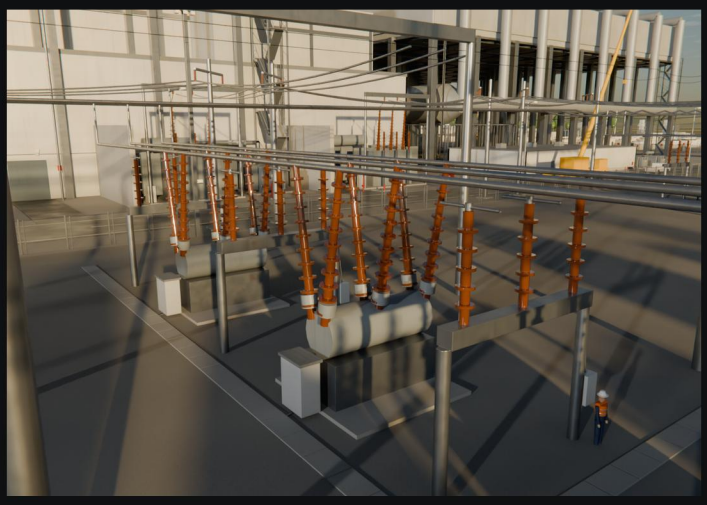
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THE 230 kV SWITCHYARD · SIX BREAKER-AND-A-HALF DIAMETERS



Along the switchyard at eye level



Bay D2: breakers, disconnects, trench to the relay house

WHAT RUNS AT THE GRID CONNECTION

HV XLPE 230 kV

1/C 2500 KCMIL CU · XLPE · CU WIRE SHIELD · AL LAMINATE · HDPE

Underground 230 kV circuits and the CCS / modular-yard ties

BARE COPPER

4/0 AWG GROUND GRID · RODS · RISERS

Station ground grid under the yard and every structure

TYPE TC-ER CONTROL

600 V MULTICONDUCTOR · EXPOSED-RUN RATED

Breaker and relay circuits in the yard trenches

MV-105 35 kV

1/C 500 KCMIL AL · EPR 133% · 1/3 CN · LLDPE

Station service and collector feeders

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ARMOR-X IN THE TRAY

Three H-class gas turbines and a steam turbine share one hall. 15 kV ARMOR-X MC-HL and MV-105 run from the 13.8 kV switchgear across the tray stack to every auxiliary drive, with 600 V ARMOR-X and Type TC-ER for power and control in the gas-turbine enclosures.

01 HAZARDOUS-LOCATION RATED
MC-HL: continuously welded, corrugated aluminium armor under a PVC jacket, for the Class I, Div. 2 areas around the gas turbines.

02 NO CONDUIT
Armored cable goes straight into the tray: fewer trades, faster pulls.

03 READABLE EVERYWHERE
Printed legend on the jacket for identification at every drop.



The legend on every jacket: ARMOR-X MC-HL 15 kV beside MV-105

15 kV MV-105 133% · RED PVC
600 V CSA RA90-HL · BLACK PVC
UL 1569 / UL 2225
TYPE TC-ER
MV-105 · UL 1072



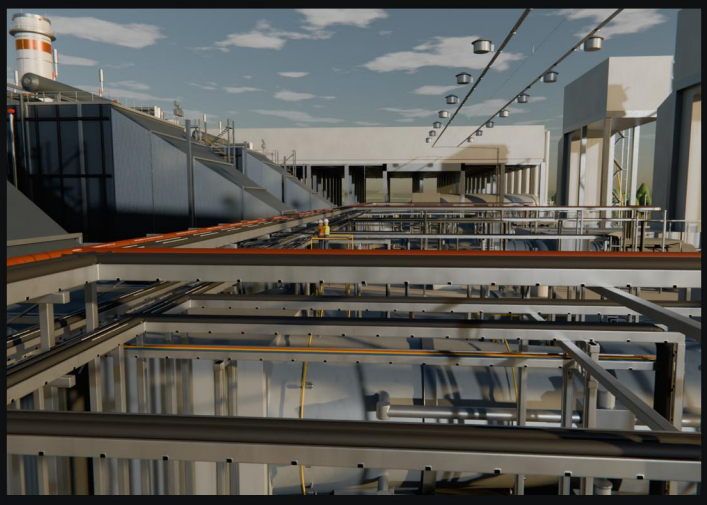
WHERE IN THE PLANT



OUTAGE PULL · NEW 13.8 kV FEEDER FROM THE LAYDOWN BAY UP INTO THE TRAY



ARMOR-X and MV-105 in the EL 48 tray over GT1



Tray stack along the hall north wall, drops to the GT skids

WHAT RUNS IN THE TURBINE HALL

ARMOR-X® MC-HL 15 kV

500 KCMIL 3/C CU · NL-EPR 25% TAPE SHIELD · GROUND · RED PVC

MV feeders to auxiliary drives through the gas-turbine areas

ARMOR-X® MC-HL 600 V

#14 AWG TO 750 KCMIL · XHHW-2 · CSA RA90-HL · BLACK PVC

Motor and power circuits on the enclosures and skids

MV-105 15 kV 133%

1/C 500 KCMIL CU · EPR · TAPE SHIELD · PVC · UL 1072

Switchgear to transformers and large motors

TYPE TC-ER

600 V POWER AND CONTROL · EXPOSED-RUN RATED

Control, protection and instrument circuits along the tray

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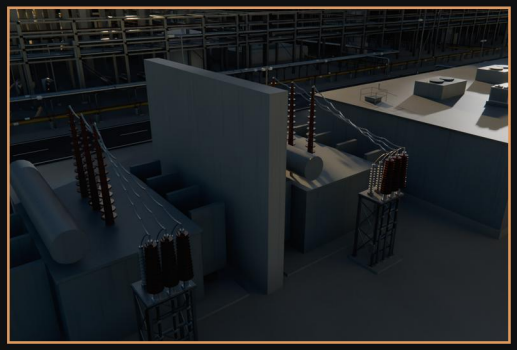
CABLE THAT RUNS CCS

Absorbers, strippers and 16 MW booster fans: post-combustion capture adds a second plant's worth of motors, drives and instruments, all fed from the CCS rack.

01 ARMOR-X ON THE RACK
Jacketed armor for the process area's power circuits.

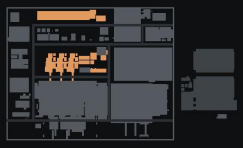
02 15 KV FEEDERS
MV-105 to the fans and the CO2 compressors.

03 INSTRUMENTATION
PLTC and Type TC-ER for the absorber and regeneration loops.



CCS transformers T-1 / T-2: 230 kV terminations and the tray riser

230 kV XLPE | 15 kV MV-105 | 600 V ARMOR-X | TYPE TC-ER | PLTC



WHERE IN THE PLANT

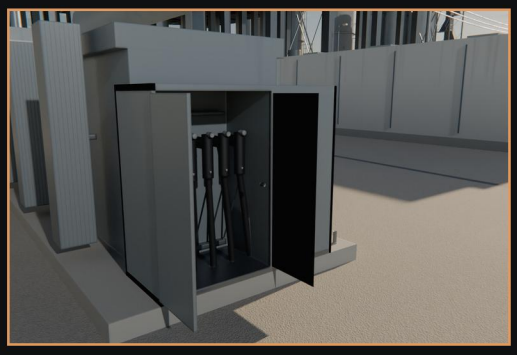
DC IN THE TRENCH

The battery yard pairs containers with power-conversion skids. 1,500 V DC conductors run in precast trenches; 35 kV MV-105 collects the skids to the BESS substation.

01 PV / RHW-2
Sunlight- and moisture-resistant 2 kV conductors for the DC side.

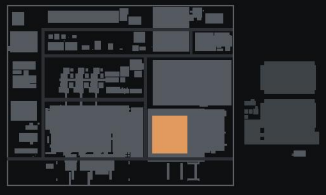
02 MV COLLECTION
35 kV MV-105 loop feeds with load-break elbows.

03 PULLED ON SITE
Crews pay off from reels straight into the open trench.



PCS / MV skid: 34.5 kV elbow terminations to the collector

2 kV PV / RHW-2 | UL 4703 | 35 kV MV-105 | UL 1072



WHERE IN THE PLANT



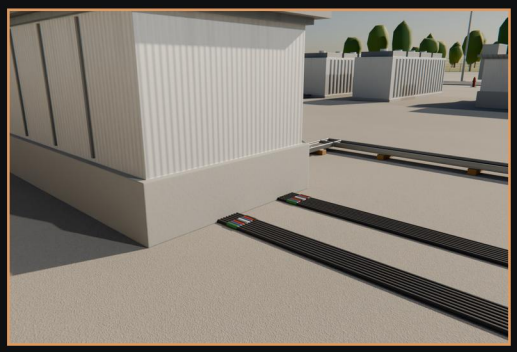
POWER THAT MOVES

Gensets, trailer turbines and bridge-power fleets come and go. Portable power cable lies on the pad, crosses lanes under drive-over protectors and plugs in with cam-locks; bridge units feed through ground trays to the permanent duct bank.

01 TYPE SHD-GC
Flexible 15 kV portable power cable for the MV feeders.

02 TYPE W
2000 V single conductors with cam-lock connections for 480 V hookups.

03 MV-105 LEADS
Generator leads from the TM2500 terminal boxes in ground trays.



Type W sets with cam-lock plugs into the 480 V switchboard

15 kV SHD-GC | ICEA S-75-381 | 2000 V TYPE W | CAM-LOCK | 15 kV MV-105



WHERE IN THE PLANT

ACCELERATING TIME TO POWER™

Schedule is decided in the field. The SIMpull Truck delivers multi-conductor SIMpull Reels and pays them off at the pull, so a feeder goes in with one setup instead of three.

01 SIMPULL REELS
Paralleled conductors on one reel: one pull per feeder.

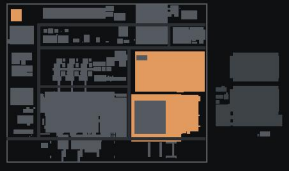
02 SIMPULL TRUCK
Delivery, payoff and reel return in one vehicle.

03 FEWER SETUPS
Less reel handling on site, faster energization.



The cable reel yard: SIMpull Reels staged for the pulls

SIMPULL® REELS | SIMPULL TRUCK | JOBSITE SERVICES



WHERE IN THE PLANT



PRODUCT INDEX

ZONE	PRODUCT	VOLTAGE	STANDARD
GRID CONNECTION	HV XLPE transmission cable	230 kV	ICEA S-108-720
	Bare copper grounding	4/0 AWG	—
POWER ISLAND	ARMOR-X® MC-HL, red PVC	15 kV	UL 1569 / 2225
	ARMOR-X® MC-HL, black PVC, CSA RA90-HL	600 V	UL 1569 / 2225
	MV-105 EPR 133%	15 kV	UL 1072
	Type TC-ER power and control	600 V	UL 1277
CARBON CAPTURE	ARMOR-X, MV-105, TC-ER, PLTC	300 V – 15 kV	UL 13 / 1072 / 1277
BATTERY STORAGE	PV / RHW-2 DC conductor	2 kV	UL 4703
	MV-105 collection	35 kV	UL 1072
MODULAR POWER	Type SHD-GC portable power cable	15 kV	ICEA S-75-381
	Type W portable power cable	2000 V	UL 1650
INSTALLATION	SIMpull® Reels and SIMpull Truck	—	—

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POWER GENERATION SOLUTIONS

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Renders are conceptual illustrations from the SK-3X1 3D model: not engineered, not for construction. Product constructions shown on cable jackets and in this index are typical; confirm ratings, sizes and listings against current Southwire specifications before use. ARMOR-X and SIMpull are registered trademarks of Southwire Company, LLC.

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